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$$\int \frac{x+2}{x^2+4x+7} dx$$
$$= \int \frac{x+2}{(x+2)^2+3} dx$$

Stel $u = x + 2 \Rightarrow du = dx$

$$= \int \frac{u}{u^2+3} du$$

Stel $t = u^2 + 3 \Rightarrow dt = 2udu \Rightarrow \frac{1}{2}dt = udu$

$$= \frac{1}{2} \int \frac{dt}{t}$$

$$= \frac{1}{2} \ln |t| + C = \frac{1}{2} \ln |u^2 + 3| + C$$

$$= \frac{1}{2} \ln |(x+2)^2 + 3| + C$$

$$= \frac{1}{2} \ln |x^2 + 4x + 7| + C$$

$$= \frac{1}{2} \ln(x^2 + 4x + 7) + C$$

References